TERM MOCK EXAM

Answers

Check yourself after you finish. Part 2 shows the model answer.

PART 1 · MULTIPLE CHOICE

Q1	D	Q2	D	Q3	B	Q4	B
Q5	A	Q6	C	Q7	B	Q8	C
Q9	D	Q10	C	Q11	C	Q12	B
Q13	A	Q14	A	Q15	B	Q16	D
Q17	D						

PART 2 · MODEL ANSWERS

Q21

1. $2x(x^2 + 4x - 3) = 2x^3 + 8x^2 - 6x$
2. $-1(x^2 + 4x - 3) = -x^2 - 4x + 3$
3. Add: $2x^3 + 7x^2 - 10x + 3$

Q22

1. $x^2 - 10x + 24$: -4 and -6 → $(x - 4)(x - 6)$
2. $x^2 + x - 20$: 5 and -4 → $(x + 5)(x - 4)$

Q23

1. Axis $x = -\frac{2}{2(1)} = -1$
2. Vertex $f(-1) = -7$ → $(-1, -7)$, minimum
3. y-intercept -6
4. Table: $x = -3, -2, -1, 0, 1$ → $y = -3, -6, -7, -6, -3$
5. Plot and join with a smooth U-shaped curve.

Q24

1. $y = -x - 1$
2. $x^2 - 6x + 5 = -x - 1$ → $x^2 - 5x + 6 = 0$
3. $(x - 2)(x - 3) = 0$ → $x = 2$ or $x = 3$
4. $(2, -3)$ and $(3, -4)$. Check $(2, -3)$: $4 - 12 + 5 = -3$ ✓, $2 + (-3) = -1$ ✓